P1

$$\frac{\partial F_{\lambda\mu}}{\partial x^{\nu}} + \frac{\partial F_{\mu\nu}}{\partial x^{\lambda}} + \frac{\partial F_{\nu\lambda}}{\partial x^{\mu}} = 0$$ All 3 indices must be different otherwise LHS = 0 trivially: for example, let $\mu = \nu \Rightarrow$

$$\frac{\partial F_{\lambda\mu}}{\partial x^{\mu}} + \frac{\partial F_{\mu\mu}^{=0}}{\partial x^{\lambda}} + \frac{\partial F_{\mu\lambda}}{\partial x^{\mu}} =$$

$$= \frac{\partial}{\partial x^{\mu}}\left(F_{\lambda\mu} + F_{\mu\lambda}\right) = 0.$$

So, let $\lambda = 0, \mu = 1, \nu = 2$:

$$\frac{\partial F_{01}}{\partial x^{2}} + \frac{\partial F_{12}}{\partial x^{0}} + \frac{\partial F_{20}}{\partial x^{1}} = 0$$

$$\frac{\partial (E_x/c)}{\partial y} + \frac{\partial(-B_z)}{\partial(ct)} + \frac{\partial}{\partial x}\left(-\frac{E_y}{c}\right) = 0$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} + \frac{\partial B_z}{\partial t} = 0$$

$$\left(\vec{\nabla} \times \vec{E}\right)_z = -\frac{\partial B_z}{\partial t}$$

$\Rightarrow$ this eqn. gives us $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$$\frac{\partial H^{\lambda\mu}}{\partial x^{\mu}} = 0$$

$\lambda = 0$: $\quad -c\,\vec{\nabla} \cdot \vec{D} = 0$

$\lambda = 1$: $\quad \frac{\partial (cD_x)}{c\,\partial t} + \frac{\partial}{\partial y}\left(-H_z\right) + \frac{\partial}{\partial z} H_y = 0$

$$\frac{\partial}{\partial y} H_z - \frac{\partial H_y}{\partial z} = \frac{\partial D_x}{\partial t} \Rightarrow \left(\vec{\nabla} \times \vec{H}\right)_x = \left(\frac{\partial \vec{D}}{\partial t}\right)_x$$

(b) $J^\alpha = (c\rho, \vec{J})$

Expect $\dfrac{\partial H^{\lambda\mu}}{\partial x^\mu} = a J^\lambda$ $\Rightarrow$ need to find a

$\lambda = 0$: $-c\vec{\nabla}\cdot\vec{D} = a J^0 = a c \rho$

$\lambda = 1$:

$$\frac{c\,\partial D_x}{c\,\partial t} - \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right) = a J^1 = a J_x$$

$$\left(\vec{\nabla}\times\vec{H}\right)_x = \frac{\partial D_x}{\partial t} - a J_x = \left(\frac{\partial \vec{D}}{\partial t} - a\vec{J}\right)_x$$

Must be: $\left.\begin{aligned} \vec{\nabla}\cdot\vec{D} &= \rho \\ \vec{\nabla}\times\vec{H} &= \vec{J} + \frac{\partial \vec{D}}{\partial t} \end{aligned}\right\} \Rightarrow a = -1.$

$$\Rightarrow \frac{\partial H^{\lambda\mu}}{\partial x^\mu} = -J^\lambda$$

(c) $u^\mu = \gamma(1, \vec{v}/c)$, $\gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}$

In the rest frame: $F^{\mu\nu}u_\nu = \begin{cases} \frac{1}{c}E_\mu & \mu = 1,2,3 \quad x\ y\ z \\ 0 & \mu = 0 \end{cases}$

$u^\mu = (1, 0)$

$$H^{\mu\nu}u_\nu = \begin{cases} c D_\mu & \mu = 1,2,3 \\ 0 & \mu = 0 \end{cases}$$

$$\vec{D} = \varepsilon\vec{E} \Rightarrow c D_\mu = \varepsilon c^2\left(\frac{1}{c}E_\mu\right)$$

Thus $H^{\mu\nu}u_\nu = c^2\varepsilon F^{\mu\nu}u_\nu$ (1)

$F_{\lambda\mu} u_\nu + F_{\mu\nu} u_\lambda + F_{\nu\lambda} u_\mu$ must have all indices different

$\lambda = 0, \mu = 1, \nu = 2$:

$$F_{01} \underset{0}{\underset{\parallel}{u_2}} + F_{12} \underset{1}{\underset{\parallel}{u_0}} + F_{20} \underset{0}{\underset{\parallel}{u_1}} = -B_z$$

$$\left(H_{\lambda\mu} u_\nu + H_{\mu\nu} u_\lambda + H_{\nu\lambda} u_\mu \right)_{\substack{\lambda=0 \\ \mu=1 \\ \nu=2}} = -H_z$$

$\vec{B} = \mu \vec{H} \Rightarrow$ the relation is:

$$(2) \quad \left(F_{\lambda\mu} u_\nu + F_{\mu\nu} u_\lambda + F_{\nu\lambda} u_\mu \right) = \mu \left(H_{\lambda\mu} u_\nu + H_{\mu\nu} u_\lambda + H_{\nu\lambda} u_\mu \right)$$

(d) Now apply eqns (1) and (2) in a moving frame, where $u_\nu = \gamma (1, -\vec{v}/c)$

<u>Eq. 2</u> Here $\underset{\frac{E_x}{c}}{\underset{\parallel}{F_{01}}} \underset{-\gamma \frac{v_y}{c}}{\underset{\parallel}{u_2}} + \underset{-B_z}{\underset{\parallel}{F_{12}}} \underset{\gamma}{\underset{\parallel}{u_0}} + \underset{-\frac{E_y}{c}}{\underset{\parallel}{F_{20}}} \underset{-\gamma \frac{v_x}{c}}{\underset{\parallel}{u_1}} =$

$$= \gamma \left(-B_z + \frac{1}{c^2} (v_x E_y - v_y E_x) \right) = \gamma \left(-B_z + \frac{1}{c^2} (\vec{v} \times \vec{E})_z \right)$$

$$H_{01} u_2 + H_{12} u_0 + H_{20} u_1 =$$

$$= \gamma \left(-H_z + c D_x (-v_y/c) + (-c D_y)(-v_x/c) \right) =$$

$$= \gamma \left(-H_z + v_x D_y - v_y D_x \right) = \gamma \left(-H_z + (\vec{v} \times \vec{D})_z \right)$$

So we have $-\gamma\left(\vec{B} - \frac{1}{c^2}\vec{v}\times\vec{E}\right)_z = \mu(-\gamma)\left(\vec{H} - \vec{v}\times\vec{D}\right)_z$

$$\boxed{\vec{B} - \frac{1}{c^2}\vec{v}\times\vec{E} = \mu\left(\vec{H} - \vec{v}\times\vec{D}\right).}$$

Eq 1

$$H^{\mu\nu}u_\nu\Big|_{\mu=0} = -c\vec{D}\cdot\left(-\frac{\gamma}{c}\vec{v}\right) = \gamma\,\vec{D}\cdot\vec{v}$$

$$H^{\mu\nu}u_\nu\Big|_{\mu=1} = cD_x\gamma - H_z\left(-\frac{\gamma}{c}v_y\right) + H_y\left(-\frac{\gamma}{c}v_z\right) =$$

$$= \gamma\left(cD_x + \frac{1}{c}\left(\vec{v}\times\vec{H}\right)_z\right)$$

$$F^{\mu\nu}u_\nu\Big|_{\mu=0} = -\frac{1}{c}\vec{E}\cdot\left(-\frac{\gamma}{c}\vec{v}\right) = \frac{\gamma}{c^2}\vec{E}\cdot\vec{v}$$

$$F^{\mu\nu}u_\nu\Big|_{\mu=1} = \frac{1}{c}E_x\gamma + \frac{\gamma}{c}B_z v_y - \frac{\gamma}{c}B_y v_z =$$

$$= \frac{\gamma}{c}\left(E_x + \left(\vec{v}\times\vec{B}\right)_x\right)$$

From $H^{\mu\nu}u_\nu = c^2\varepsilon F^{\mu\nu}u_\nu$ follows:

$$\gamma\,\vec{D}\cdot\vec{v} = c^2\varepsilon\,\frac{\gamma}{c^2}\vec{E}\cdot\vec{v} = \varepsilon\gamma\,\vec{E}\cdot\vec{v}$$

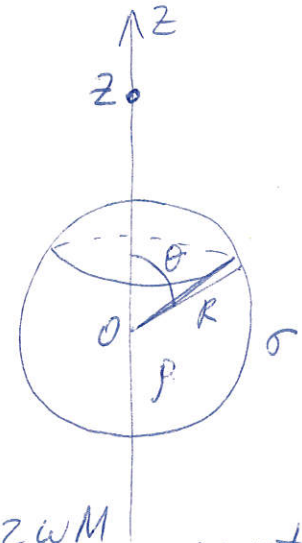
$$\Rightarrow \vec{D}\cdot\vec{v} = \varepsilon\,\vec{E}\cdot\vec{v} \quad \text{i.e.} \quad \vec{D}_{\parallel} = \varepsilon\,\vec{E}_{\parallel}$$ along axis of motion.

$$\gamma\left(c\vec{D} + \frac{1}{c}\vec{v}\times\vec{H}\right) = c^2\varepsilon\,\frac{\gamma}{c}\left(\vec{E} + \vec{v}\times\vec{B}\right)$$

$$\boxed{\vec{D} + \frac{1}{c^2}\vec{v}\times\vec{H} = \varepsilon\left(\vec{E} + \vec{v}\times\vec{B}\right)}$$

P2 Uniformly magnetized $\vec{M} = M\hat{z}$ sphere rotating with $\vec{\omega} = \omega\hat{z}$ around magnetization axis. lecture note shows that, for $\varepsilon \approx \varepsilon_0$,

$$\begin{cases} \rho_{bulk} = -\dfrac{2\omega M}{c^2} \\ \sigma_{surf} = \dfrac{\omega M}{c^2} R \sin^2\theta \end{cases}$$



Since $\rho = -\dfrac{2\omega M}{c^2} = \text{const}$, the field it produces in that of point charge $Q_b = \rho \dfrac{4\pi}{3} R^3 = -\dfrac{8\pi}{3c^2}\omega M R^3$

$$\vec{E}_b = \frac{Q_b \hat{z}}{4\pi\varepsilon_0 z^2} = -\frac{2\omega M R^3}{3\varepsilon_0 c^2 z^2}\hat{z}$$

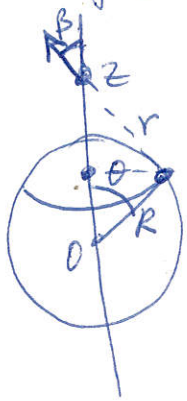
The surface contribution is (next page)

$$\vec{E}_s = \frac{2\omega M R^3}{3\varepsilon_0 c^2}\left(\frac{1}{z^2} - \frac{3R^2}{5z^4}\right)\hat{z}$$

$$\Rightarrow \text{total } \vec{E}(z) = -\frac{2\omega M R^5}{5\varepsilon_0 c^2 z^4}\hat{z}$$

This shows that the charge distribution possesses quadrupole moment so that $E(z) \sim \frac{1}{z^4}$.

Electric field due to surface charge σ_{surf}:



$$r^2 = (z - R\cos\theta)^2 + (R\sin\theta)^2 =$$

$$= z^2 + R^2 - 2zR\cos\theta$$

$$\cos\beta = \frac{z - R\cos\theta}{r}$$

$$E_S = \frac{1}{4\pi\varepsilon_0} \int_0^{2\pi} d\varphi \int_0^{\pi} \underbrace{\sigma_{surf}(\theta) \frac{R\sin\theta\, R d\theta}{r^2}}_{\text{field charge of surface element } R\sin\theta \cdot R d\theta \cdot d\varphi} \underbrace{\frac{z - R\cos\theta}{r}}_{\cos\beta \text{ (projection on } \hat{z}\text{-axis)}}$$

$$= \frac{\omega M R^3}{2\varepsilon_0 c^2} \int_0^{\pi} d\theta \sin\theta \sin^2\theta \frac{z - R\cos\theta}{(z^2 + R^2 - 2zR\cos\theta)^{3/2}} =$$

$x = \cos\theta$

$$= \frac{\omega M R^3}{2\varepsilon_0 c^2} \underbrace{\int_{-1}^{1} dx \frac{(z - Rx)(1 - x^2)}{(z^2 + R^2 - 2zRx)^{3/2}}}_{I}$$

$$E_z = \frac{\omega M R^3}{2\varepsilon_0 c^2} \int_{-1}^{1} dx \, \frac{(z - Rx)(1-x^2)}{(z^2+R^2-2zRx)^{3/2}} =$$

$$= \frac{\omega M R^3}{2\varepsilon_0 c^2}\left(\frac{4}{3z^2} - \frac{4R^2}{5z^4}\right) = \frac{2\omega M R^3}{3\varepsilon_0 c^2}\left(\frac{1}{z^2} - \frac{3R^2}{5z^4}\right)$$

This is how it is calculated:

Let $y = z^2+R^2-2zRx$, $x = \frac{z^2+R^2-y}{2zR}$

$dy = -2zR\,dx$

$$I = \int_{(z+R)^2}^{(z-R)^2} \frac{-dy}{2zR} \frac{\left(z - \frac{z^2+R^2-y}{2z}\right)\left(1 - \frac{(z^2+R^2-y)^2}{4z^2R^2}\right)}{y^{3/2}} =$$

$$= \frac{1}{2zR}\int_{(z-R)^2}^{(z+R)^2} dy \, \frac{\frac{z^2-R^2+y}{2z} \, \frac{4z^2R^2-(z^2+R^2)^2-y^2+2y(z^2+R^2)}{4z^2R^2}}{y^{3/2}}$$

$$= \frac{-R}{(2zR)^4}\int_{(z-R)^2}^{(z+R)^2} \frac{dy}{y^{3/2}} \left(z^2-R^2+y\right)\left(y^2 - 2y(z^2+R^2) + (z^2-R^2)^2\right)$$

$$= \frac{-R}{(2zR)^4}\int \frac{dy}{y^{3/2}} \Big\{ y^3 + y^2\left[z^2-R^2-2(z^2+R^2)\right] +$$

$$+ y\left[(z^2-R^2)^2 - 2(z^2-R^2)(z^2+R^2)\right] + (z^2-R^2)^3 \Big\}$$

Now: $\int_{(z-R)^2}^{(z+R)^2} dy \, y^{3/2} = \frac{2}{5} y^{5/2}\Big|_{(z-R)^2}^{(z+R)^2} = \frac{2}{5}\left[(z+R)^5 - (z-R)^5\right] =$

$\underline{z-R>0}$

$$= \frac{2}{5}\cdot 2\cdot\left(R^5 + 10R^3z^2 + 5Rz^4\right)$$

$$\int dy\, y^{1/2} = \frac{2}{3}\, y^{3/2}\Big|_{(z-R)^2}^{(z+R)^2} = \frac{2}{3}\left((z+R)^3 - (z-R)^3\right)$$
$$= \frac{4}{3}\left(R^3 + 3Rz^2\right)$$

$$\int \frac{dy}{y^{1/2}} = 2\, y^{1/2}\Big|_{(z-R)^2}^{(z+R)^2} = 2\left((z+R) - (z-R)\right) = 4R$$

$$\sqrt{(z-R)^2} = |z-R| = z-R \text{ for } z>R$$
(point outside sphere)

$$\int \frac{dy}{y^{3/2}} = -2\cdot\frac{1}{y^{1/2}}\Big|_{(z-R)^2}^{(z+R)^2} = -2\left(\frac{1}{z+R} - \frac{1}{z-R}\right) = \frac{4R}{z^2-R^2}.$$

$$I = \frac{-R}{(2zR)^4}\left\{\frac{4}{5}\left(R^5 + 10R^3z^2 + 5Rz^4\right) - \left(z^2+3R^2\right)\cdot\right.$$
$$\cdot\frac{4}{3}\left(R^3+3Rz^2\right) - \left(z^2-R^2\right)\left(z^2+3R^2\right)\cdot 4R$$
$$\left. + \left(z^2-R^2\right)^3\cdot\frac{4R}{z^2-R^2}\right\} = \frac{-R}{(2zR)^4}\cdot$$

$$\cdot\left\{\left(z^2-R^2\right)4R\left[z^2-R^2-\left(z^2+3R^2\right)\right] - \frac{4R}{3}\left(R^2+3z^2\right)\left(z^2+3R^2\right)\right.$$
$$\left. + \frac{4R}{5}\left(R^4+10R^2z^2+5z^4\right)\right\} = \frac{-R}{(2zR)^4}\cdot$$

$$\cdot\left\{-16R^3\left(z^2-R^2\right) - \frac{4R}{3}\left(3R^4+3z^4+10R^2z^2\right)\right.$$
$$\left. + \frac{4R}{5}\left(R^4+10R^2z^2+5z^4\right)\right\} =$$

$$= \frac{-R}{(2zR)^4}\left\{-16R^3z^2 + 16R^5 - 4R^5 - \cancel{4Rz^4} - \frac{40}{3}R^3z^2\right.$$
$$\left. + \frac{4}{5}R^5 + 8R^3z^2 + \cancel{4Rz^4}\right\} = \frac{-1}{16z^4R^3}\cdot$$

$$\cdot\left\{\frac{64}{5}R^5 - \frac{64}{3}R^3z^2\right\} = \frac{4}{3z^2} - \frac{4R^2}{5z^4}$$

Prob.3

Problem 2. The equation for the drift velocities are

$$\frac{d\vec{v}_1}{dt} = -\frac{e}{m_1}\vec{E} - \frac{e}{m_1}\vec{v}_1 \times \vec{B} - \frac{\vec{v}_1}{\tau} = 0,$$

$$\frac{d\vec{v}_2}{dt} = \frac{e}{m_2}\vec{E} + \frac{e}{m_2}\vec{v}_2 \times \vec{B} - \frac{\vec{v}_2}{\tau} = 0,$$

each equation is solved independently, similar to how it was done in class, then $\vec{j} = -en_1\vec{v}_1 + en_2\vec{v}_2$

The total conductivity is the sum of the two conductivities

$$\sigma_{ik} = \sigma_{ik}^{(1)} + \sigma_{ik}^{(2)}$$

where $$\sigma_{ik}^{(1)} = \frac{e^2 n_1 \tau_1}{(\omega_{c1}^2\tau_1^2 + 1)m_1}\begin{pmatrix} 1 & -\omega_{c1}\tau \\ \omega_{c1}\tau & 1 \end{pmatrix}$$

where $\omega_{c1} = \frac{eB}{m_1} > 0$

Similarly,

$$\sigma_{ik}^{(2)} = \frac{e^2 n_2 \tau_2}{(\omega_{c2}^2 \tau^2 + 1) m_2} \begin{pmatrix} 1 & \omega_{c2}\tau \\ -\omega_{c2}\tau & 1 \end{pmatrix} \qquad \omega_{c2} = \frac{eB}{m_2} > 0$$

The Hall current (perpendicular to both $\vec{E}$ and $\vec{B}$) is given by off-diagonal terms

$$j_H = \frac{-e^2 n_1 \tau_1^2 E \omega_{c1}}{(\omega_{c1}^2 \tau^2 + 1) m_1} + \frac{e^2 n_2 \tau_2^2 \omega_{c2} E}{(\omega_{c2}^2 \tau^2 + 1) m_2}$$

For strong magnetic field, such that

$$\omega_{c1}\tau \gg 1 \quad \text{and} \quad \omega_{c2}\tau \gg 1$$

the Hall current simplifies to

$$j_H = -\frac{e^2 n_1}{m_1 \omega_{c1}} E + \frac{e^2 n_2}{m_2 \omega_{c2}} = \frac{e(n_2 - n_1)E}{B}$$

We see that for equal number of electrons and holes (compensated semiconductor) the Hall effect vanishes. Note that for n-type and p-type semiconductors the Hall current has <u>opposite</u> sign.